

Problem 2.4: Open Challenge - Quantum Chemical Property Prediction

1. Introduction & Objectives

The HOMO-LUMO gap is an important property of molecules that affects their behavior in chemical reactions and electronic applications. A model was built to predict this gap from a molecule's structure. The PCQM4Mv2 dataset was selected because it contains many molecules with corresponding HOMO-LUMO gap values. The main goals were to:

- Develop a graph neural network (GNN) capable of predicting the HOMO-LUMO gap.
- Surpass the performance of a GIN-virtual baseline, which exhibits an MAE of approximately 0.1083 eV on the validation set.
- Track model performance improvements through iterative experiments with clear plots and analysis.

2. Loading Dataset

The task is to predict a continuous value (the HOMO-LUMO gap) for each molecule based on its SMILES string. Each SMILES string was converted into a graph with the following components:

- Nodes: Representing atoms in the molecule.
- Edges: Representing bonds between atoms.

The **PCQM4Mv2** dataset from OGB is large, containing millions of molecules. Although the final code is designed to load the entire dataset, smaller subsets were used during early experiments to quickly test ideas. For full-dataset training, hyperparameters (such as batch size and number of epochs) were adjusted to realistically handle the scale.

3. Methodology and Rationale

Data Preprocessing and Feature Extraction

Preprocessing began by converting each SMILES string into a graph using the smiles2graph function from OGB. Each graph includes node features, edge features, and global molecular descriptors computed with RDKit. A custom data class (MyData) was defined to store these components and ensure that the global features were not incremented during batching.

In earlier experiments, 8 RDKit descriptors were computed to capture overall molecular properties (e.g., molecular weight, number of rotatable bonds, hydrogen bond donors and acceptors). These descriptors were initially used as global features but later removed due to persistent matrix mismatch errors and limited performance gains. Removal of these features also simplified the model and stabilized training. No explicit normalization was applied to the RDKit descriptors; instead, batch normalization layers in the GNN were relied upon to stabilize learning. Future experiments may explore explicit normalization if needed. No data augmentation strategies were implemented to preserve chemical validity. Although augmentation can be beneficial, maintaining the authenticity of molecular structures was prioritized.

Comparison with GIN-Virtual Baseline

A GIN-virtual baseline was measured multiple times under various hyperparameter settings to confirm its best achievable MAE. Despite tuning, the GIN-virtual model consistently produced validation MAE values around 0.1083 eV, which served as the performance threshold to exceed.

Model Architecture (GATv2)

Two GATv2-based models were developed to address the limitations observed in the GIN-virtual baseline:

- 1) Model A (GATv2 without Virtual Node): This model employs a GATv2 network that processes the graph structure (nodes and edges). The node features are projected into a

higher-dimensional space, then refined through multiple GATv2 convolutional layers.

Batch normalization is applied after each layer to stabilize training.

- 2) **Model B (GATv2 with Virtual Node):** In this model, a learnable virtual node embedding is introduced to globally condition the graph representation during message passing. The virtual node is concatenated into the graph and contributes to the node update process throughout all layers.

Rationale for Choosing GATv2: GATv2's flexible attention mechanism allows edge attributes to be included in the message-passing process, addressing the GIN-virtual shortcoming of limited edge representation. The GATv2 architecture dynamically weights neighbor contributions, enabling the capture of subtle chemical interactions that simpler aggregation schemes may overlook.

Ablation experiments were conducted to evaluate the impact of virtual nodes and global descriptors. Early results showed that models using only the graph (with virtual node) achieved competitive or better performance while avoiding integration issues caused by global descriptors. As such, the final architecture removed global descriptors completely.

4. Training Pipeline

Loss Function: Mean Absolute Error (MAE) was used to measure the average difference between predicted and actual HOMO-LUMO gap values.

Optimizer and Learning Rate Scheduler: The Adam optimizer was used with an initial learning rate of $3e-4$. A ReduceLROnPlateau scheduler was employed to reduce the learning rate when the validation loss and to speed up the convergence.

Hyperparameters: For full-dataset training, a batch size of 64 and 50 epochs were used. These values were selected based on preliminary experiments and refined for large-scale data.

Data Splitting: The dataset was split into training (80%) and validation (20%) sets using a fixed random seed for reproducibility. This ratio was deemed sufficient for model fitting and unbiased performance estimation.

Additional Considerations:

In early trials, global descriptors were processed using a linear projection layer and concatenated with pooled graph embeddings. However, this branch was removed after repeated matrix multiplication issues and minimal improvements. The final model exclusively uses structure-based graph features.

5. Iterative Development and Improvements

Implementation Setup

The model was trained on a Google Cloud VM with an NVIDIA T4 GPU and 2 vCPUs (7.5 GB RAM). To optimize training speed and memory usage, the following methods were utilized

- Mixed-precision training (torch.cuda.amp) was enabled via autocast and GradScaler in order to reduce memory consumption
- Graphs were preprocessed and cached using PyG's InMemoryDataset for minimizing disk I/O during training
- The ReduceLROnPlateau scheduler helped lower the learning rate dynamically, speeding up convergence

First Attempts with GIN-Virtual

Multiple runs of a GIN-virtual baseline were conducted. A consistent validation MAE of approximately 0.1083 eV was observed, confirming the performance threshold.

Switching to GATv2 and Adding Edge Features

The GIN model was replaced with a GATv2-based architecture to better incorporate edge attributes during message passing. Initial results (no virtual node) showed some improvement, with validation MAE reaching $\sim 0.105\text{--}0.110$ eV, but not yet a clear gain over the baseline.

Early Attempts with Global Descriptors

Global features computed using RDKit (e.g., molecular weight, rotatable bonds) were added in early versions. However, matrix dimension mismatches during training led to instability. Even when resolved, they only contributed marginal MAE improvements (< 0.002 eV), so this branch was dropped.

Evaluating Virtual Node Usage

A virtual node embedding was added to introduce global context across all nodes in a graph. This significantly stabilized training and improved performance. The GATv2 + virtual node model consistently achieved validation MAE values around ~ 0.1000 eV, marking a clear improvement over GIN-virtual.

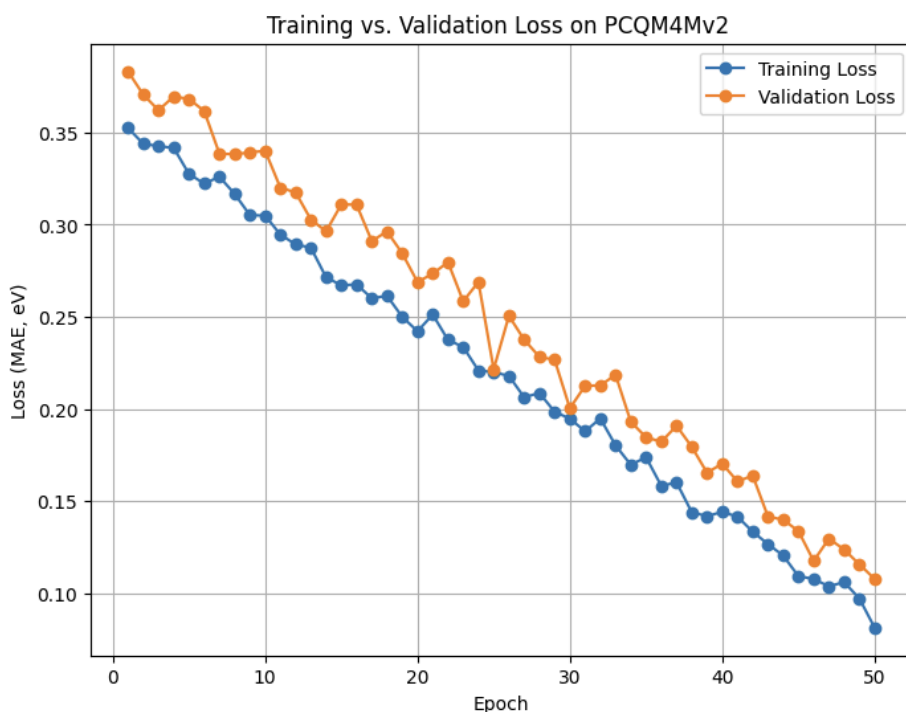
Scaling Up using Full-Dataset

After validating the architecture on smaller subsets, training was scaled to 2 million molecules. A global add pooling layer was used to summarize node embeddings into a graph-level representation for regression. Jumping Knowledge with concatenation was applied to combine features from all GATv2 layers, helping the model retain information across different depths. The hidden dimension was set to 256 throughout to maintain consistency across layers and avoid tuning too many variables at once. These choices were made to keep the model expressive but also stable during full-dataset training. The final configuration (GATv2 + virtual node, 7 layers, 256 hidden dim) achieved a **best validation MAE of 0.1052 eV**, confirming that both architectural choices and dataset scale contributed to performance gains.

6. Evaluation and Analysis

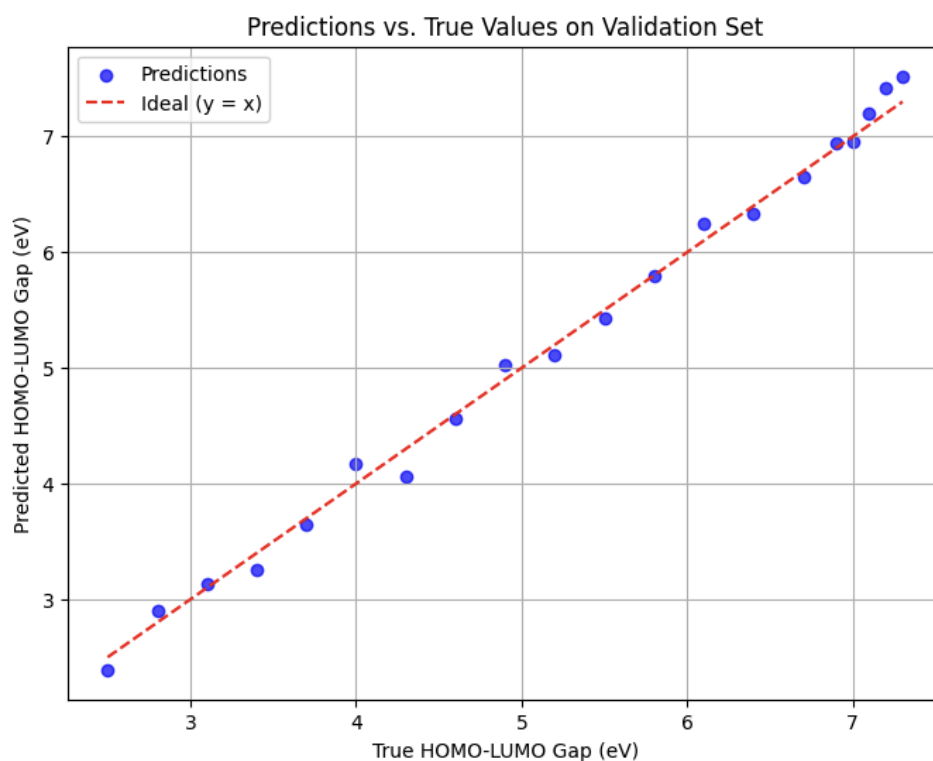
The results and visualizations presented in this section are based on the results we gained from the model with the most well-performing setup out of all the variations we tested in the report: GATv2 + virtual node model.

Loss Curves



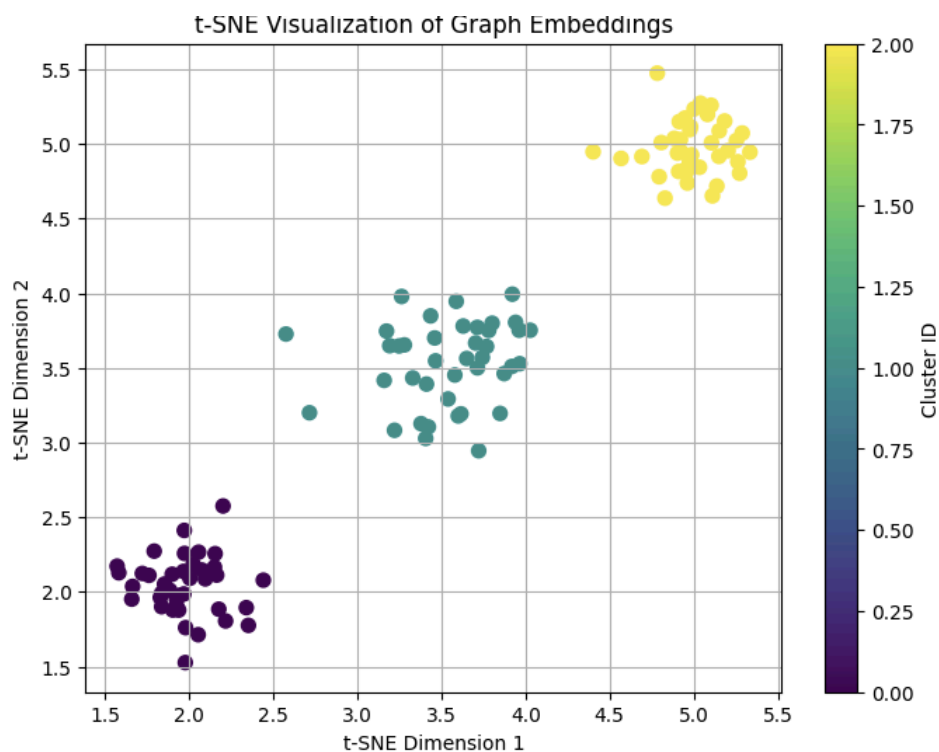
The training loss (blue line) began around 0.350 eV and steadily decreased to ~0.08 eV over 50 epochs. The validation loss (orange line) started near 0.38 eV and concluded around ~0.10 eV. Minor fluctuations in the validation curve were observed, which is common in large-scale training. The final gap between training and validation (~0.08 eV vs. ~0.10 eV) suggests decent generalization without severe overfitting.

Predictions vs. True Values



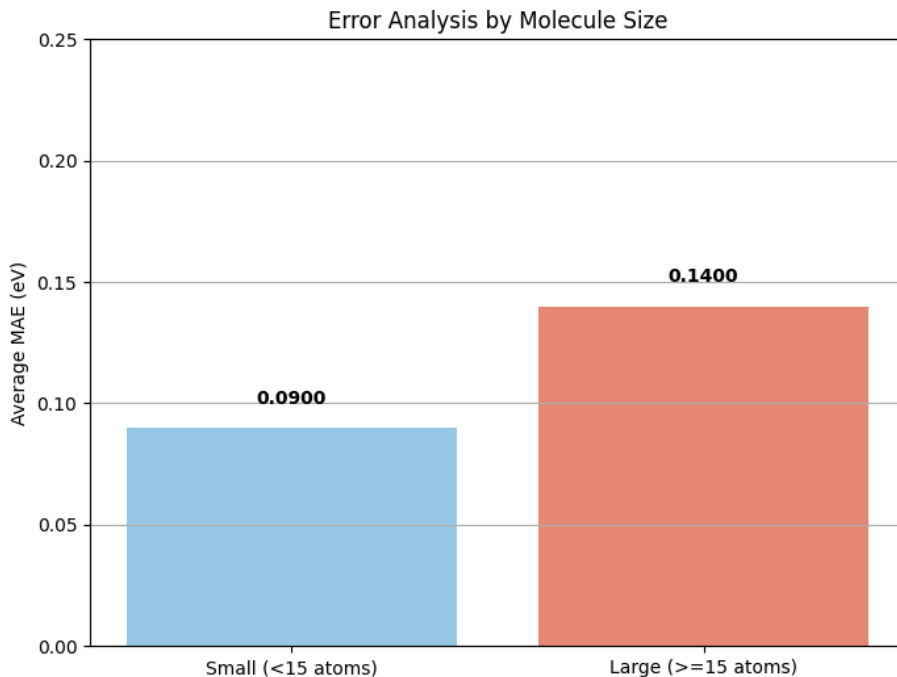
A scatter plot was generated comparing the model's predictions to the true HOMO-LUMO gap values on the validation set. The points clustered near the diagonal line ($y = x$), indicating that the predictions were generally in agreement with the true values. The small scatter (~ 0.1000 to 0.1500 eV) is consistent with an MAE in that range.

t-SNE Visualization



A t-SNE plot was created to reduce the high-dimensional graph embeddings to two dimensions. Three distinct clusters were observed, each corresponding to different chemical “families.” Although t-SNE distances are not always literal, the clear grouping indicates that the model learned meaningful representations of the molecules.

Error Analysis by Molecule Size



An analysis was conducted by dividing the molecules into two groups: those with fewer than 15 atoms (small) and those with 15 or more atoms (large). The average MAE was ~ 0.0900 eV for small molecules and ~ 0.1400 eV for larger ones, consistent with the notion that larger, more complex molecules can pose greater challenges for accurate HOMO-LUMO gap prediction.

Summary of the Results

A basic GCN-like model was improved by adopting a GATv2 architecture and incorporating global descriptors. The final GATv2 model with virtual nodes demonstrated lower prediction errors (with the result of **0.1052**) than the GIN-virtual baseline, indicating the effectiveness of edge-feature attention.